

B.Sc. (Part-II) Fourth Semester (NEP) Summer 2026  
109207/110207/112207/123207/134207/135207 - Major - Computer  
Science/Computer Application(R)/Computer Application (Voc.)/Data  
Analytics/Information Technology/Data Science and Analytics - OOP with C++

P. Pages : 1

Time : Two Hours



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Max. Marks : 30

- Notes :
1. All questions are compulsory.
  2. Draw neat diagrams wherever necessary.
  3. Assume suitable data wherever necessary.
  4. Illustrate your answer with the help of neat sketches wherever necessary.

1. Write answers in one or two lines solve **any six**.

6

- i) What is Object-Oriented Programming?
- ii) What is abstraction?
- iii) What is constructor?
- iv) What is function overriding?
- v) What is a base class?
- vi) What is early binding?
- vii) What is Inline function
- viii) What is multiple inheritance?

2. Answers in short **any two** of the following (3 marks each)

6

- a) What is OOP? Explain the features of OOP.
- b) Explain the advantages of Object Oriented Programming.
- c) What is data abstraction and encapsulation? Explain.
- d) How to define class and create objects? Explain with suitable example.

3. Explain in detail **any one** of the following.

6

- a) What is function argument? Explain default argument with suitable example
- b) What is constructor? Explain its types

4. Answer in short **any two** of the following (3 marks each)

6

- a) Explain function access specifiers with suitable example.
- b) Explain returning object from function with example.
- c) What is inheritance? Explain Single inheritance.
- d) Explain Hybrid inheritance with suitable example.

5. Explain in detail **any one** of the following.

6

- a) What is virtual function? Explain pure virtual function suitable example.
- b) What is Operator Overloading? Explain unary operator overloading in detail.

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